# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

**Syllabus: Chapter 11 — Basic Statistics**

| **Total Marks: 65** | **Time Allowed: 2h 30m** | **Version: 1** |
|---|---|---|

**General Instructions:**

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for graphs and diagrams.

## SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

*Encircle the correct option. All questions are compulsory.*

| # | Question | A | B | C | D | Answer |
|---|---|---|---|---|---|---|
| 1 | Data which has not been arranged in a systematic order is called: | Grouped data | Raw data | Cumulative data | Discrete data | ○○○○ |
| 2 | The class mark (mid value) of the class 21–30 is: | 21 | 25 | 25.5 | 30 | ○○○○ |
| 3 | For the class 31–40, the lower class boundary is: | 31 | 30 | 30.5 | 31.5 | ○○○○ |
| 4 | Cumulative frequency of a class is the sum of its frequency and the frequencies of all: | Succeeding classes | Preceding classes | All classes equally | The modal class only | ○○○○ |
| 5 | In a histogram with unequal class intervals, the height of each bar represents: | Frequency | Class mark | Frequency density | Cumulative freq. | ○○○○ |
| 6 | Arithmetic mean of the data: 2, 4, 6, 8, 10 is: | 5 | 6 | 7 | 8 | ○○○○ |
| 7 | For grouped data the arithmetic mean using class marks is: | Σx / n | Σfx / Σf | Σf / Σx | Σfx × n | ○○○○ |
| 8 | The median of: 3, 5, 7, 9, 11 is: | 5 | 7 | 8 | 9 | ○○○○ |
| 9 | In Mode = l + $[(f_m-f_1)/(2f_m-f_1-f_2)]\times h$, 'l' is: | Class mark of modal class | Lower boundary of modal class | Upper boundary of modal class | Lower limit of preceding class | ○○○○ |
| 10 | If mode = 26 and mean = 23, the median by empirical relation is: | 21 | 24 | 25 | 27 | ○○○○ |
| 11 | The weighted arithmetic mean formula is: | Σw / Σwx | Σwx / Σw | Σwx × Σw | Σx / Σw | ○○○○ |
| 12 | In a frequency polygon, points are plotted against: | Class boundaries | Frequencies only | Class marks (mid values) | Cumulative frequencies | ○○○○ |

## SECTION B — Short Questions (11 × 3 = 33 Marks)

*Attempt ALL 11 questions. Choose either the main question OR its alternative.*

| Q. | Question | Marks | OR | OR Question | Marks |
|---|---|---|---|---|---|
| 2(i) | A library data shows: 1–10 (f=5), 11–20 (f=4), 21–30 (f=8), 31–40 (f=9), 41–50 (f=2), 51–60 (f=2). State: (a) total number of readers, (b) group with highest frequency, (c) lower boundary of the last class. | 3 | OR | Define: (a) raw data, (b) class limits, (c) class boundaries. Find the adjustment factor for classes 1–10, 11–20. | 3 |
| 2(ii) | For the class interval 1–9, find: (a) lower class limit, (b) upper class limit, (c) class boundaries, (d) class mark. | 3 | OR | Explain **frequency density** and write its formula. A class has frequency 12 and class interval width 4. Find its frequency density. | 3 |
| 2(iii) | A die is rolled 30 times: 1 appears 4 times, 2 appears 6, 3 appears 5, 4 appears 5, 5 appears 4, 6 appears 6. Prepare a discrete frequency distribution table and add a cumulative frequency column. | 3 | OR | List the six steps of the **Tally Bar Method** for constructing a frequency distribution table. | 3 |
| 2(iv) | Find the arithmetic mean of x = 2, 4, 6, 8, 10, 12 using $\bar{x} = \Sigma x / n$. | 3 | OR | Find the arithmetic mean of y = 3, 4, −1, 7, −8, −5, 0. | 3 |
| 2(v) | Find the arithmetic mean of the frequency distribution:<br>x: 1, 2, 3, 4, 5<br>f: 3, 7, 5, 2, 3 | 3 | OR | Find the arithmetic mean of the fuel data:<br>Class: 0–5, 5–10, 10–15, 15–20, 20–25<br>f: 5, 7, 10, 6, 2 | 3 |

<table>
<tr><td>2(vi)</td><td>Find the median of: 40, 35, 45, 60, 50, 35, 40, 45, 35, 50.</td><td>3</td><td>OR</td><td>Selling prices of 7 articles (Rs): 1075, 1050, 2050, 1030, 2045, 1090, 1530. Find the median selling price.</td><td>3</td></tr>
<tr><td>2(vii)</td><td>Find mode and state type (unimodal/bimodal):<br>(a) 30, 25, 40, 30, 45, 25, 30, 40, 30 (b) 1, 2, 3, 2, 1, 3, 1, 2</td><td>3</td><td>OR</td><td>Find the mode of: 125, 130, 115, 125, 135, 120, 135, 130, 120.<br>State the type (unimodal / bimodal / trimodal).</td><td>3</td></tr>
<tr><td>2(viii)</td><td>Mean = 29, Median = 32. Find Mode using: Mode = 3 Median − 2 Mean.</td><td>3</td><td>OR</td><td>Mode = 26, Mean = 23. Find Median using: Median = ⅓(Mode + 2 Mean).</td><td>3</td></tr>
<tr><td>2(ix)</td><td>Find weighted mean. Marks (x): Maths 90, Physics 75, Chemistry 60, Biology 50. Weights (w): 4, 3, 2, 1 respectively.</td><td>3</td><td>OR</td><td>Define weighted mean and state its formula. A shop sells items at Rs 200 (w=5), Rs 350 (w=3), Rs 500 (w=2). Find the weighted mean price.</td><td>3</td></tr>
<tr><td>2(x)</td><td>Find median and mode:<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>f</td><td>3</td><td>4</td><td>5</td><td>2</td><td>2</td></tr></table></td><td>3</td><td>OR</td><td>Find median and mode:<table><tr><td>x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>f</td><td>2</td><td>5</td><td>7</td><td>4</td><td>2</td></tr></table></td><td>3</td></tr>
<tr><td>2(xi)</td><td>Describe how a frequency polygon is drawn on a histogram. What is done at both ends to complete a closed polygon?</td><td>3</td><td>OR</td><td>Workers’ ages: 20–24 (f=5), 25–29 (f=16), 30–34 (f=12), 35–39 (f=10), 40–44 (f=8), 45–49 (f=4). Find the class mark for each group.</td><td>3</td></tr>
</table>

## SECTION C — Long Questions (4 × 5 = 20 Marks)

*Attempt ALL 4 questions. Choose either the main question OR its alternative.*

<table>
<tr><th>Q.</th><th>Question</th><th>Marks</th><th>OR</th><th>OR Question</th><th>Marks</th></tr>
<tr><td>3(i)</td><td>Masses (pounds) of 40 boys at a gym:<table><tr><td>Mass</td><td>125–135</td><td>136–146</td><td>147–157</td><td>158–168</td><td>169–179</td></tr><tr><td>f</td><td>8</td><td>13</td><td>11</td><td>6</td><td>2</td></tr></table>(a) Find class marks for all 5 classes.<br>(b) Find class boundaries for all 5 classes.<br>(c) Construct a cumulative frequency column.</td><td>5</td><td>OR</td><td>Heights (cm) of 44 students:<table><tr><td>Ht</td><td>130–144</td><td>145–149</td><td>150–154</td><td>155–159</td><td>160–169</td><td>170–179</td></tr><tr><td>f</td><td>9</td><td>7</td><td>8</td><td>6</td><td>10</td><td>4</td></tr></table>Intervals are unequal. Calculate frequency density (F.D) and class boundaries for all 6 classes.</td><td>5</td></tr>
<tr><td>3(ii)</td><td>Find the arithmetic mean of marks (grouped data). Show full working table with class marks and fx column.<table><tr><td>Marks</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>f</td><td>2</td><td>5</td><td>7</td><td>3</td><td>8</td></tr></table></td><td>5</td><td>OR</td><td>Find the arithmetic mean (detergent packs sold). Show full working table.<table><tr><td>Class (kg)</td><td>1–9</td><td>10–18</td><td>19–27</td><td>28–36</td><td>37–45</td></tr><tr><td>f</td><td>10</td><td>15</td><td>18</td><td>15</td><td>12</td></tr></table></td><td>5</td></tr>
<tr><td>3(iii)</td><td>Marks of 100 students. Find the median marks.<table><tr><td>Marks</td><td>10-19</td><td>20-29</td><td>30-39</td><td>40-49</td><td>50-59</td></tr><tr><td>f</td><td>5</td><td>25</td><td>40</td><td>20</td><td>10</td></tr></table>Build c.f. table, identify median group, apply: $\tilde{x} = l + (h/f)(n/2 - c)$.</td><td>5</td><td>OR</td><td>Wages (Rs/hr) of 50 workers. Find the mode.<table><tr><td>Wages</td><td>25–29</td><td>30–34</td><td>35–39</td><td>40–44</td><td>45–49</td><td>50–54</td><td>55–59</td></tr><tr><td>f</td><td>2</td><td>4</td><td>8</td><td>20</td><td>6</td><td>6</td><td>4</td></tr></table>Give modal class boundaries. Apply: $\text{Mode} = l + [(f_m - f_1)/(2f_m - f_1 - f_2)] \times h$.</td><td>5</td></tr>
<tr><td>3(iv)</td><td>Monthly household expenditure with weights given below. Find the weighted arithmetic mean. Show the full wx column.<table><tr><td>Item</td><td>Food</td><td>Rent</td><td>Clothing</td><td>Utility</td><td>Education</td><td>Misc.</td></tr><tr><td>x (Rs)</td><td>8000</td><td>5000</td><td>1000</td><td>900</td><td>600</td><td>100</td></tr><tr><td>w</td><td>20</td><td>10</td><td>6</td><td>4</td><td>2</td><td>1</td></tr></table></td><td>5</td><td>OR</td><td>Marks of class 9 students with subject weights. Find weighted mean for Hanzala and Zaki; state who scores higher.<table><tr><td>Subject</td><td>w</td><td>Hanzala</td><td>Zaki</td></tr><tr><td>Mathematics</td><td>4</td><td>70</td><td>90</td></tr><tr><td>Physics</td><td>3</td><td>95</td><td>55</td></tr><tr><td>Chemistry</td><td>2</td><td>80</td><td>50</td></tr><tr><td>Biology</td><td>1</td><td>55</td><td>65</td></tr></table></td><td>5</td></tr>
</table>

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Ch 11 — Basic Statistics

**— End of Question Paper —**